

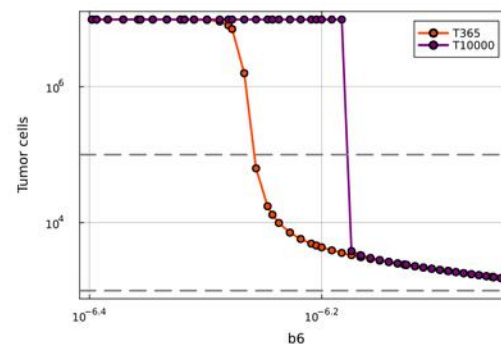
Step 0 + Step 1 only

Post-meeting feedback deck

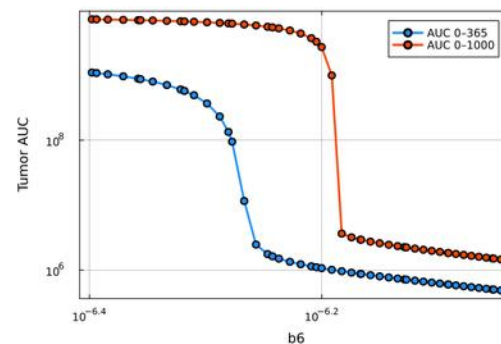
This version follows Adam's request more literally: freeze the structure first, then make the long-term dense b6 ODE result clear through trajectories and summary plots.

- Step 0: fix the simpler direct CTL formulation and stop changing model structure.
- Step 1: long-time dense 1D b6 ODE sweep with b5 fixed at $1e-4$ and baseline initial condition.
- Presentation emphasis: trajectory path first, endpoint plots second.

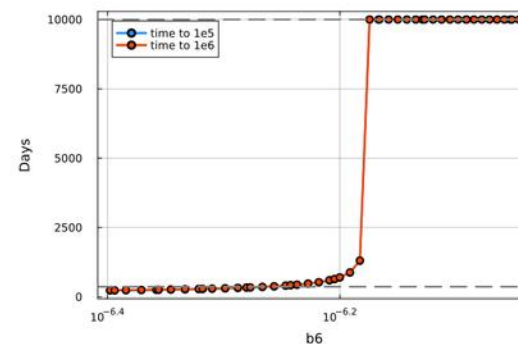
Zoomed threshold plot: T365 vs T10000



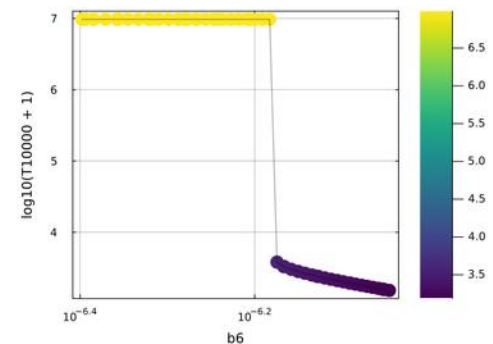
Zoomed threshold plot: early burden



Zoomed threshold plot: crossing times

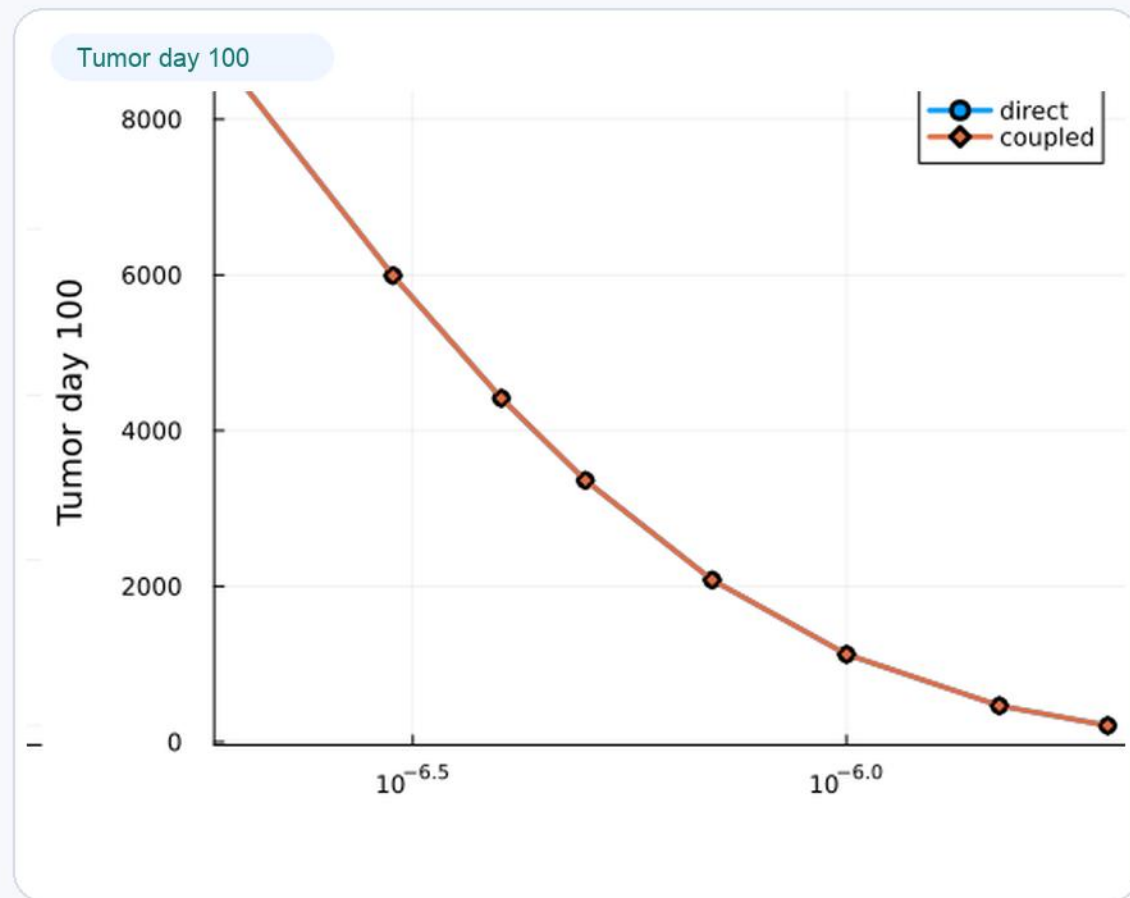


Zoomed threshold plot: long-time state



Step 0: freeze the model structure before testing robustness

Direct versus coupled CTL form near the b6 threshold



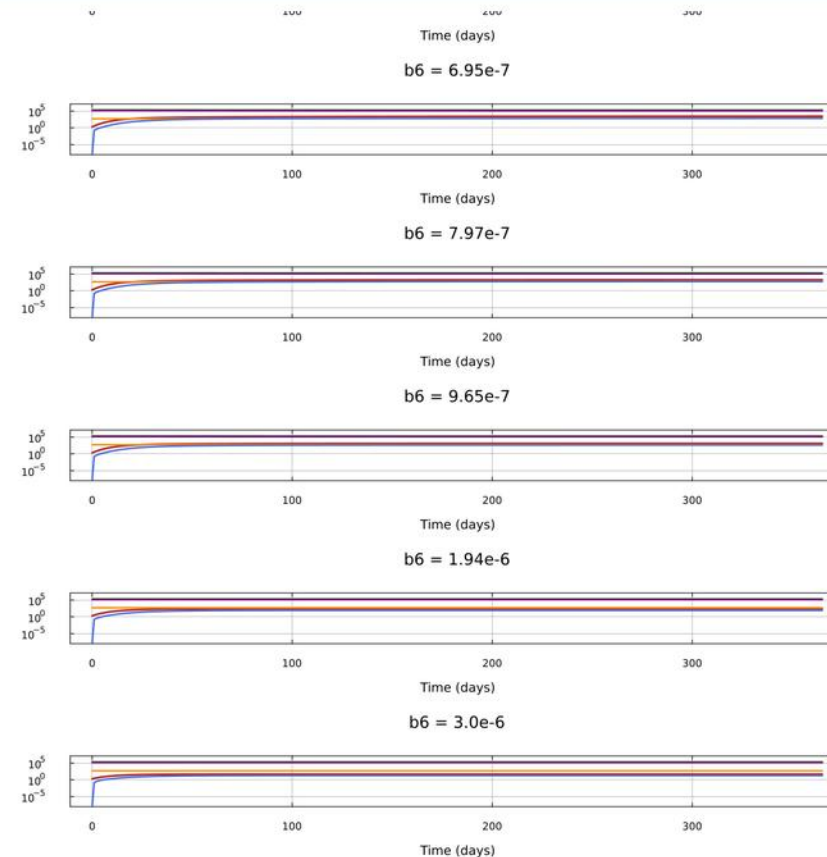
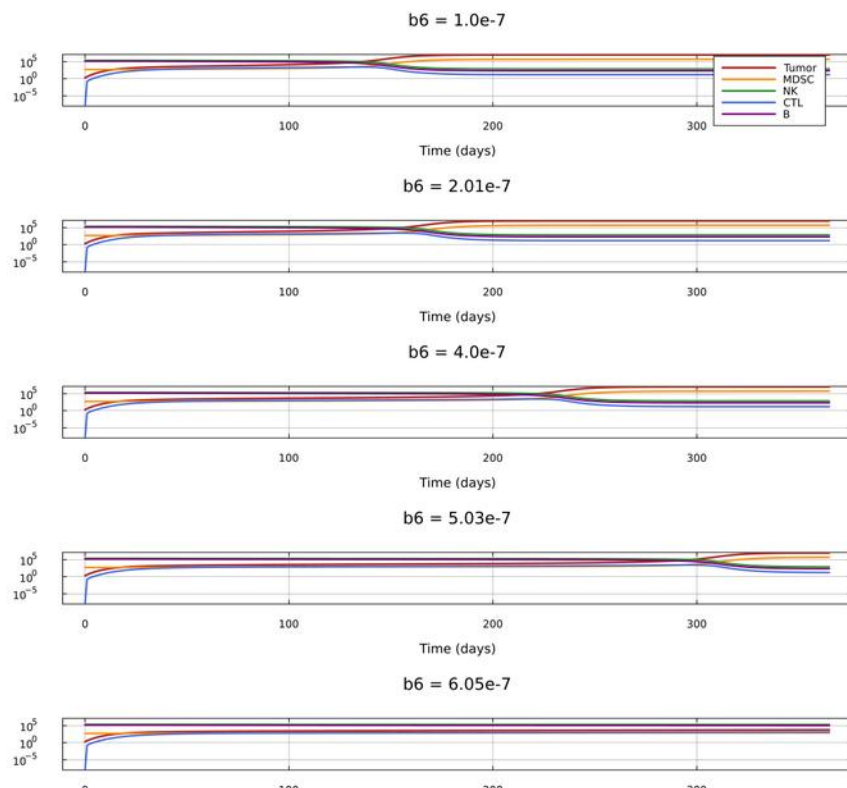
Takeaway

- Tumor day 100 and tumor day 365 are almost identical for the two CTL formulations near the threshold.
- So Step 1 fixes the simpler direct CTL formulation instead of changing structure again.
- All Step 1 runs then keep baseline initial condition and $b_5 = 1e-4$ while varying only b_6 .

Step 1A: trajectories over 0-365 days already separate by b_6

Lower b_6 cases on the left, threshold-to-high b_6 cases on the right

Representative ODE trajectories (0-365 days)



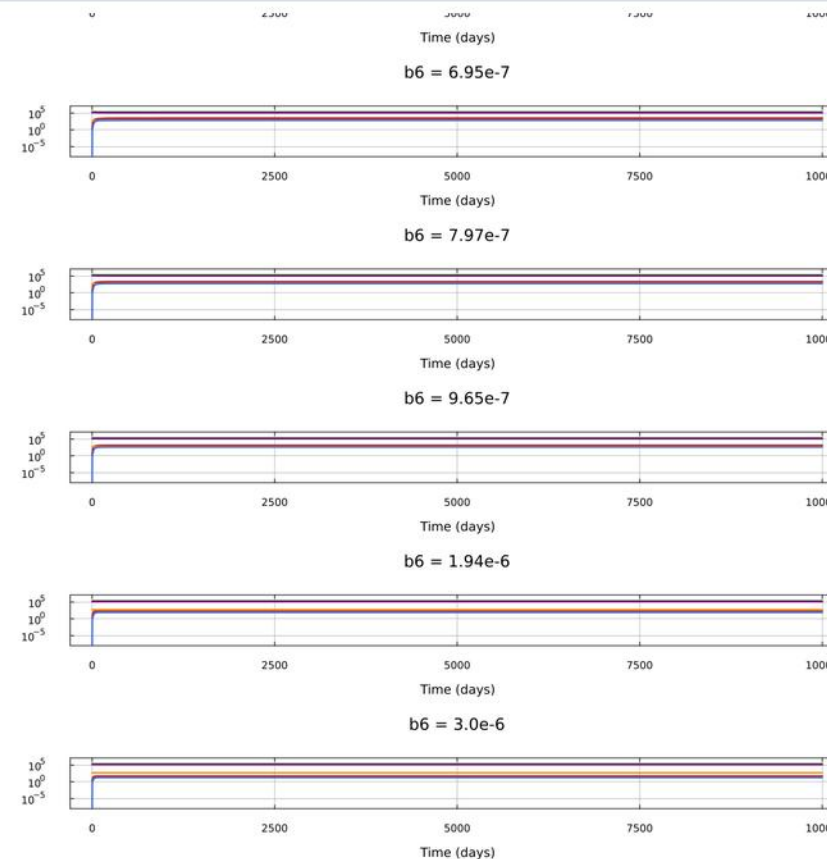
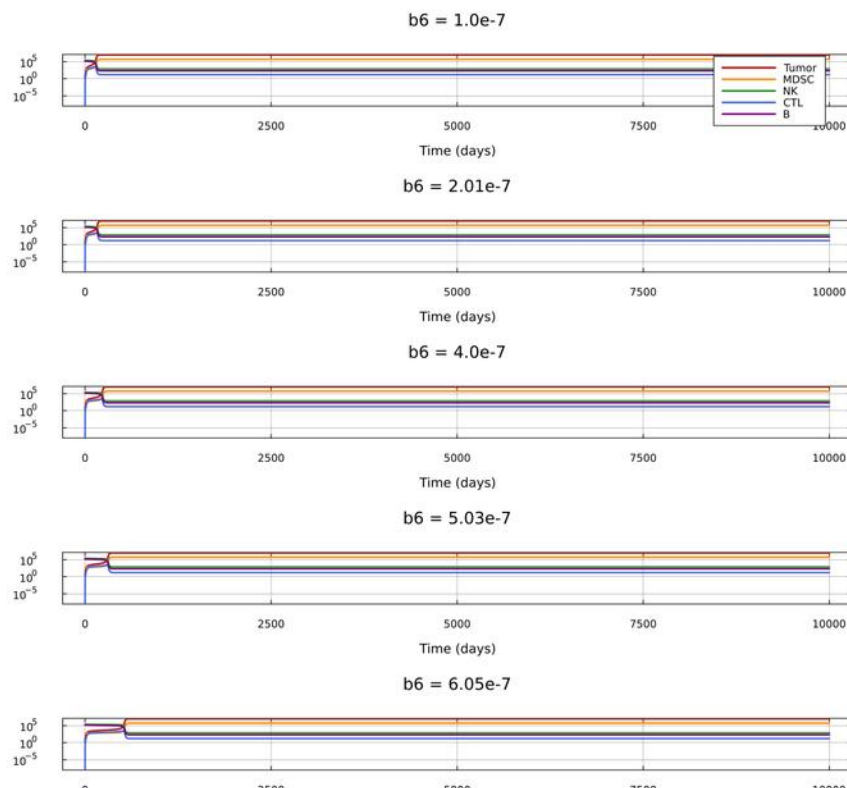
Takeaway

- Below about $6e-7$, the first-year trajectories remain clearly tumor-dominant.
- Around $7e-7$ and above, tumor drops toward the 10^3 scale by day 365.
- This is why the trajectory path matters: it shows how the response develops, not just the endpoint value.

Step 1B: by 10,000 days, delayed growth is not always durable control

Same representative b_6 values, now followed to long time

Representative ODE trajectories (0-10,000 days)



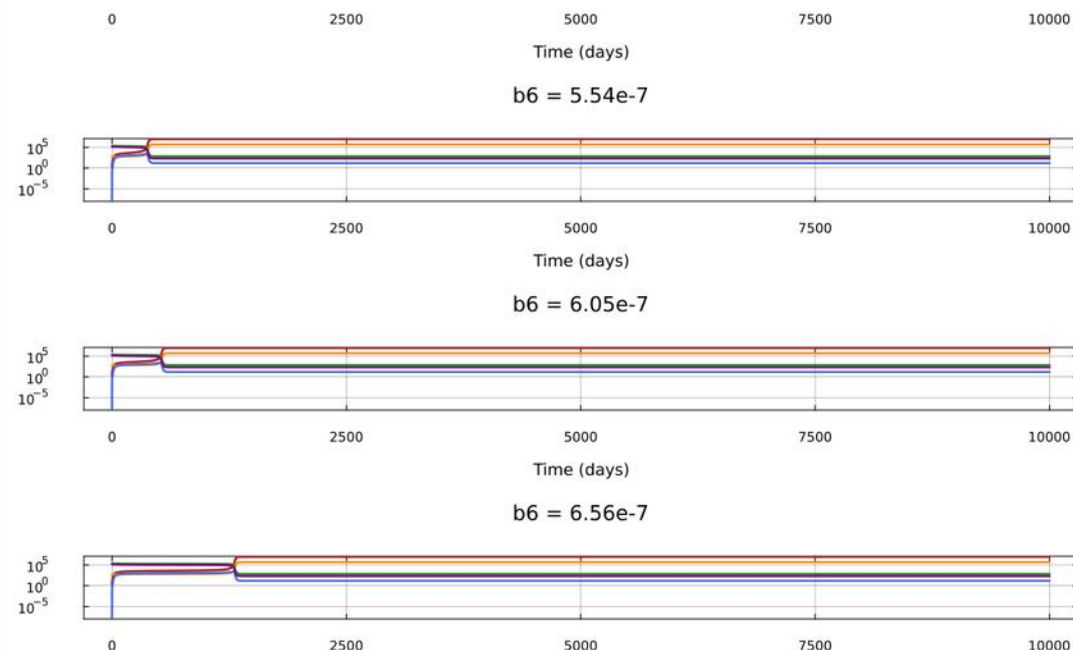
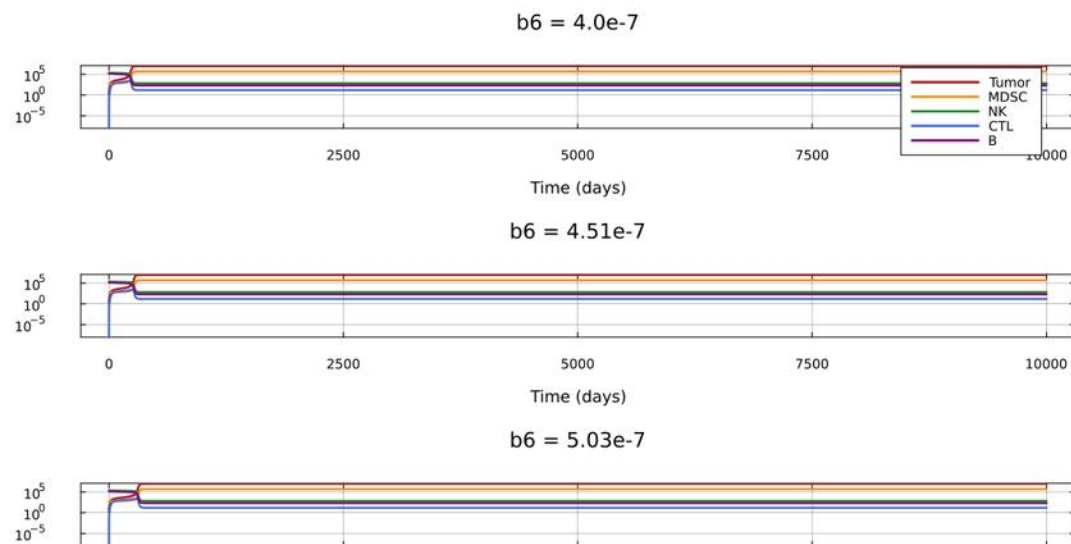
Takeaway

- The $b_6 \approx 6.05e-7$ case still escapes by long time even though it looks partially suppressed at day 365.
- The $\sim 7e-7$ region settles near an intermediate or dormancy-like state instead of rebounding to 10^7 .
- The $2e-6$ and $3e-6$ cases stay controlled through the full 10,000-day window.

Step 1C: lower side of the threshold shows delayed escape

Threshold-region trajectories, roughly $4\text{e-}7$ through $6.56\text{e-}7$

Threshold-region ODE trajectories (fine b_6 grid, 0–10,000 days)

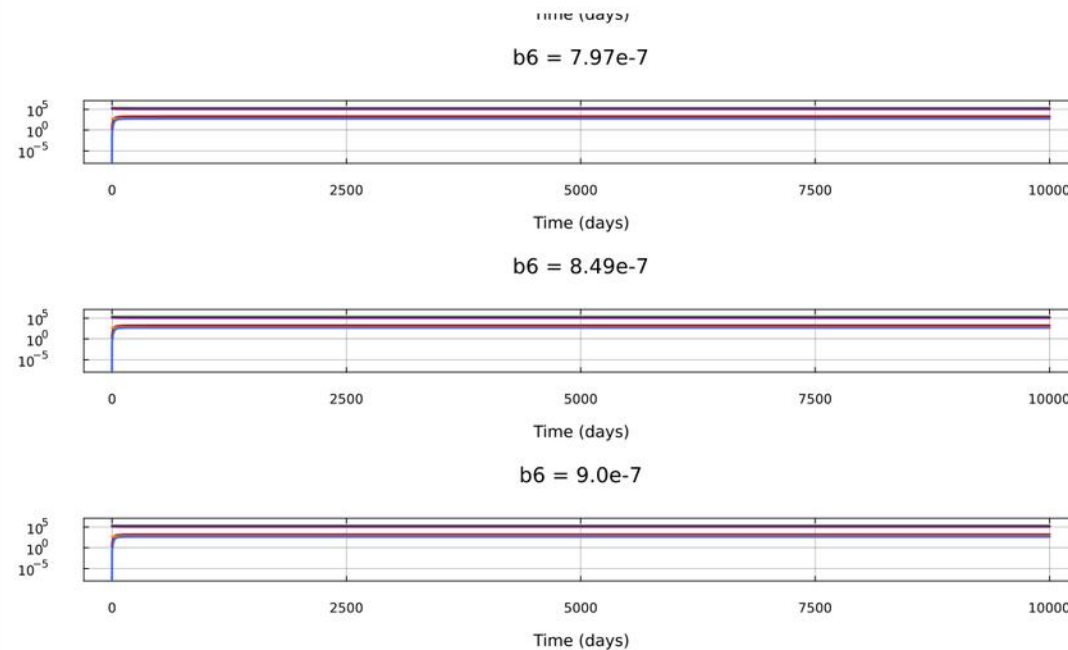
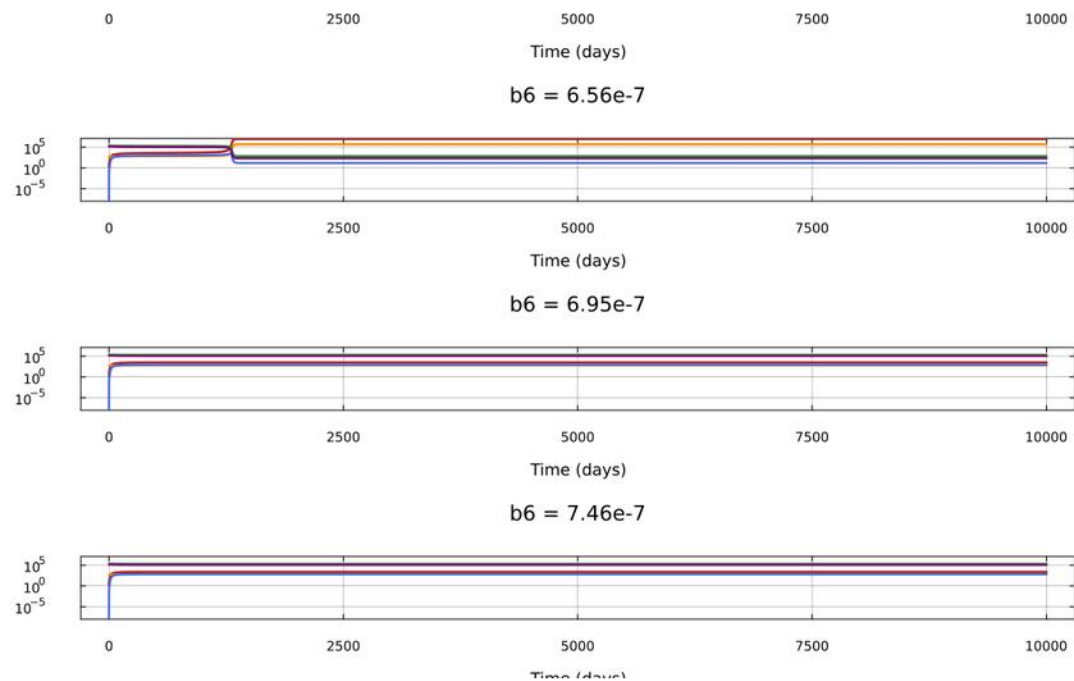


Takeaway

- From $4\text{e-}7$ to about $6.56\text{e-}7$, all cases still move toward tumor-dominant behavior.
- What changes is the escape timing: growth becomes slower and slower as b_6 increases.
- This is the pre-switch side of the result, not true long-time control yet.

Step 1D: upper side of the threshold stays low over long time

Threshold-region trajectories, roughly $6.95\text{e-}7$ through $9\text{e-}7$



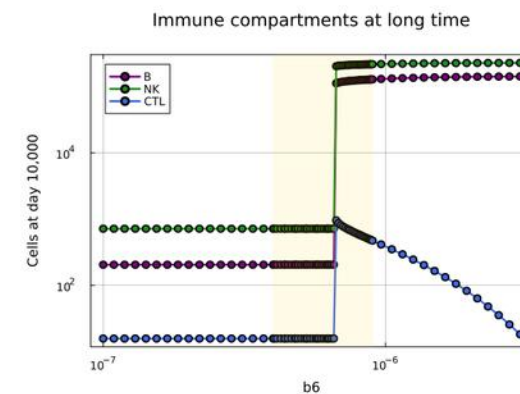
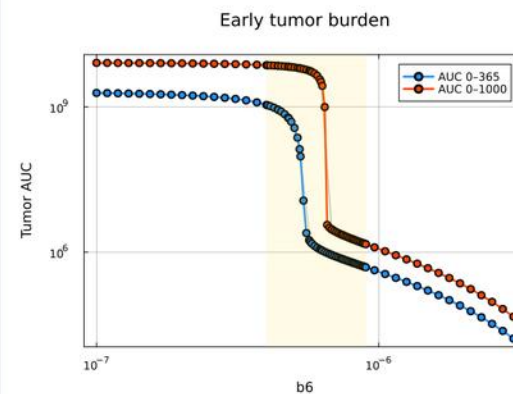
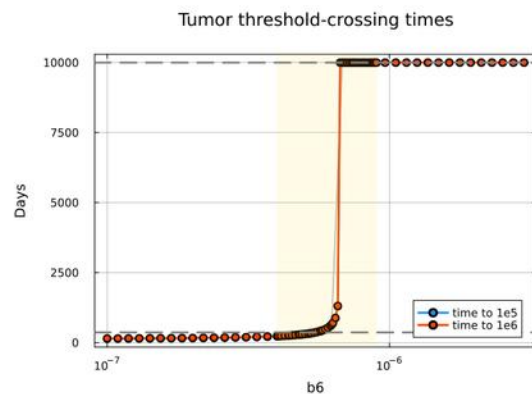
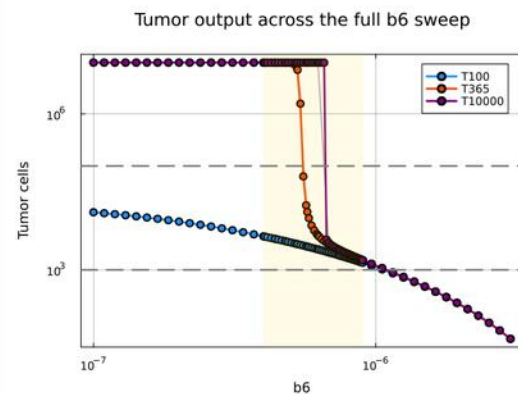
Takeaway

- Across this narrow band, the long-time tumor level stays near the 10^3 scale rather than returning to 10^7 .
- So the switch window is narrow, and it is visible directly in the trajectories.
- The real controlled regime begins later, around $1.15\text{e-}6$, outside this threshold-only figure.

This is the clearest trajectory evidence that the change is sharp, not broad.

Step 1E: ODE summary plots confirm the same story numerically

Top row: tumor outputs and crossing times; bottom row: burden and immune state

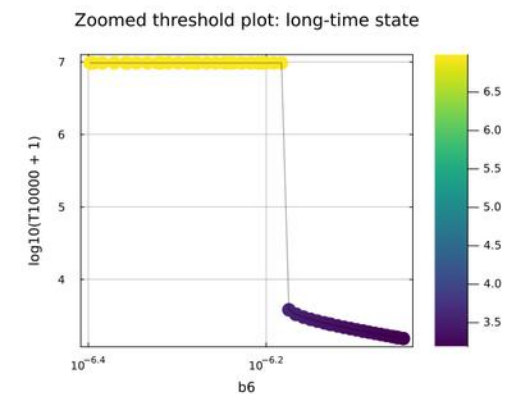
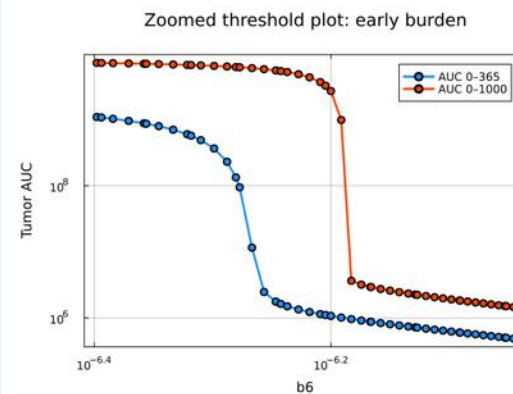
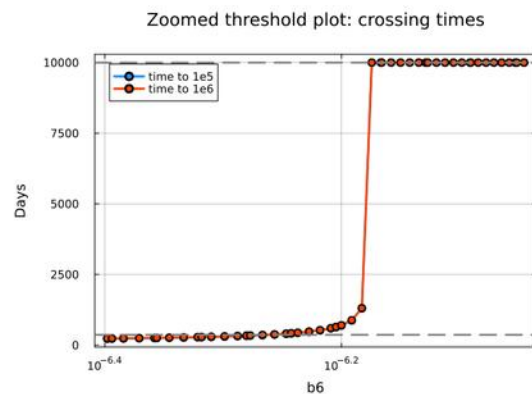
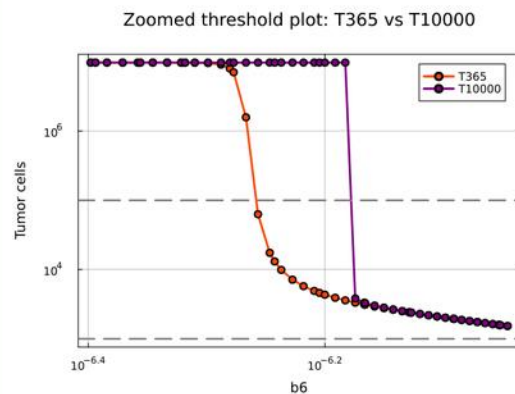


Takeaway

- T100, T365 and T10000 all move with b_6 , but the long-time readout is the strictest one.
- Time-to- $1e5$ and time-to- $1e6$ stretch sharply near the threshold window.
- The long-time control state comes with high B and high NK, while CTL peaks near the transition itself.

Step 1F: the threshold is narrow, and day 365 is not the whole story

Zoomed view of the switch region



Takeaway

- The first $T10000 < 1e5$ appears near $6.69e-7$, while the first $T10000 < 1e3$ appears near $1.15e-6$.
- T365 and T10000 do not define exactly the same threshold point.
- That is why the 10,000-day extension was necessary after the meeting feedback.

The zoomed summary turns the trajectory story into explicit threshold numbers.

How this Step 0 + Step 1 deck answers the meeting feedback

Only the current work, not future extensions

Step 0

- Freeze the structure instead of re-editing the CTL term again.
- Use the simpler direct CTL form because tumor-level outputs are unchanged near threshold.

Step 1

- Use finer b6 grades with a dense threshold region.
- Extend ODE runs to 10,000 days.
- Keep trajectory plots as a main result, not a supplement.

Current discussion points

- Is the intermediate regime biologically meaningful?
- Which endpoint should define control: day 365 or long-time behavior?
- Is the narrow b6 switch strong enough to motivate the next analysis step?